

NDA

Autumn 2023

what is bytecode? How do Java language processors work?

① a) Byte code is the intermediate code compiled and executed by a virtual machine.

Java language processors works.

① combination of compilation and interpretation.

② A Java source program is first compiled into an intermediate form called bytecodes. The bytecodes are then interpreted by a virtual machine.

b) How is an executable target program generated? Explain it using a language processing system:

⇒ The generation of an executable target program involves several steps and the process can vary depending on the programming language and the specific tools used.

generating an executable program using a language processing system:

Preprocessor ÷ A preprocessor is a tool that produces input for compilers.

Assembler ÷ Translates assembly language programs into machine code.

Linker ÷ In order to make an executable file, search and locate

referenced module.

Loader: Loader is responsible for loading executable files into memory and execute them.

cross-compiler: A compiler that runs on platform (A) and is capable of generating executable code for platform (B) is called a cross compiler.

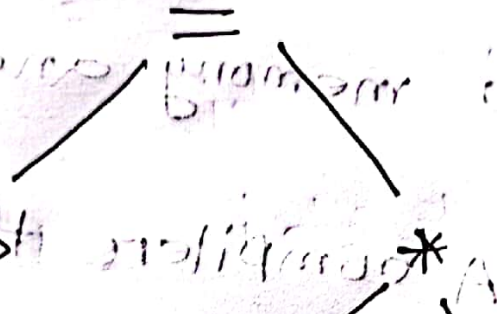
c) $z = a * b + c / 4$ [z, a, b, and c contain floating point values]

Solve

Lexical analysis

$\langle \text{id}, 1 \rangle \langle = \rangle \langle \text{id}, 2 \rangle \langle * \rangle \langle \text{id}, 3 \rangle \langle + \rangle$
 $\langle \text{id}, 4 \rangle \langle / \rangle \langle 4 \rangle$

Syntax



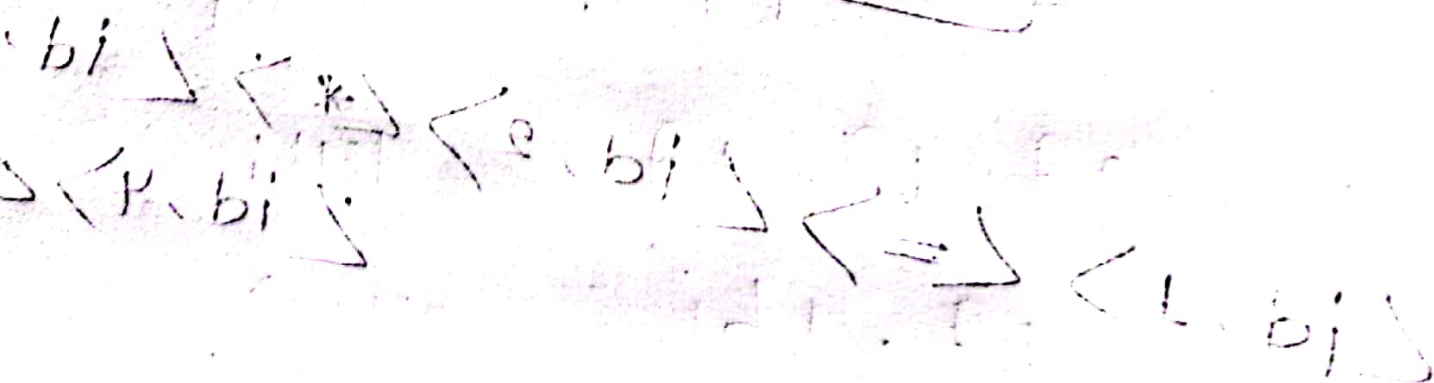
<id, 3>

<id, 4>

+

'/'

4



code optimization

$$t_1 = id_4 / 4.0$$

$$t_2 = id_3 + t_1$$

$$id_1 = id_2 * t_2$$

Target code generator

Load R3, id4

Div R3, R3, #4.0

Load (R2), id3

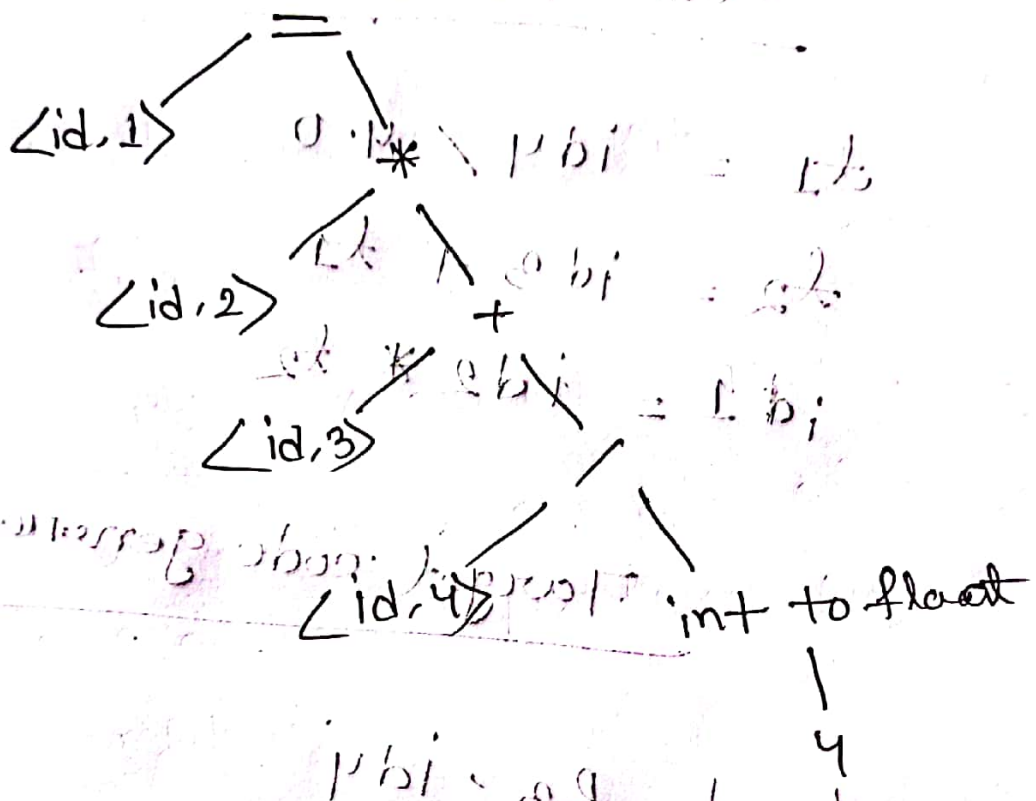
ADD R2, R2, R3

Load (R1), id2

MUL R1, R1, R2

Store id1, R1

Semantic



Intermediate code generation

$t_1 = \text{int to float}(4)$
 $t_2 = id4 \div t_1$
 $t_3 = id3 + t_2$
 $t_4 = id4 * t_3$
 ~~$t_5 = id4$~~
 $id1 = t_4$